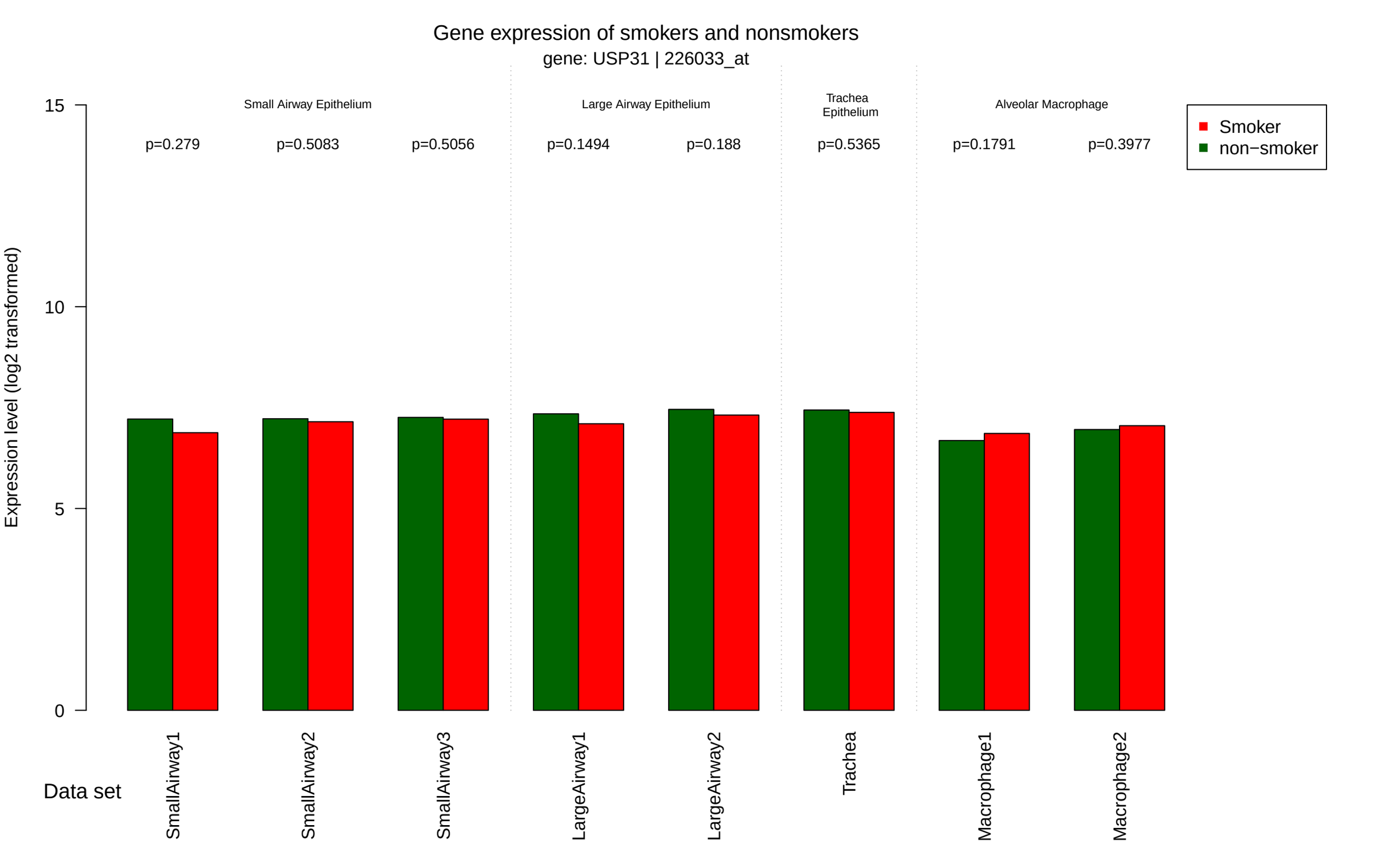
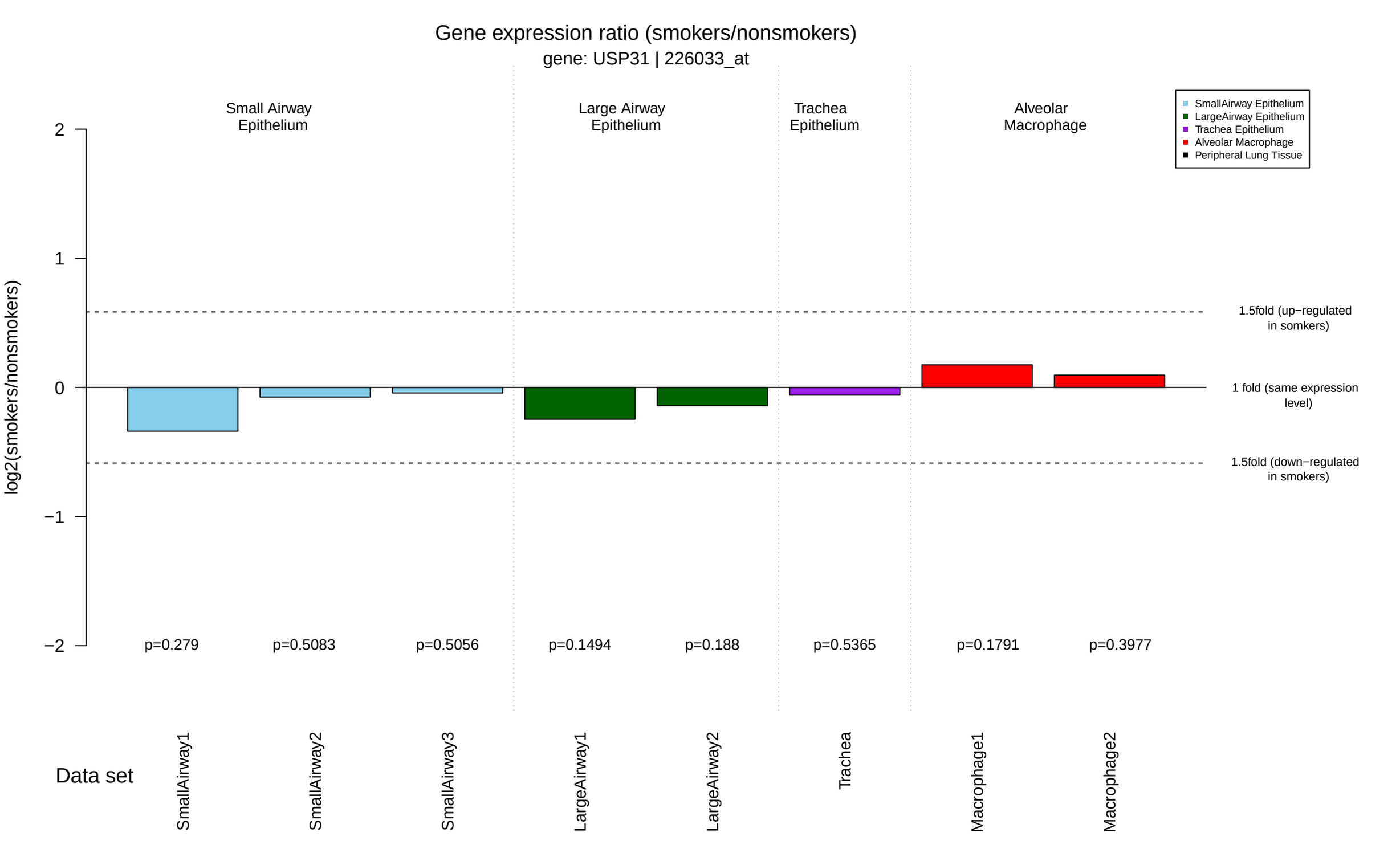
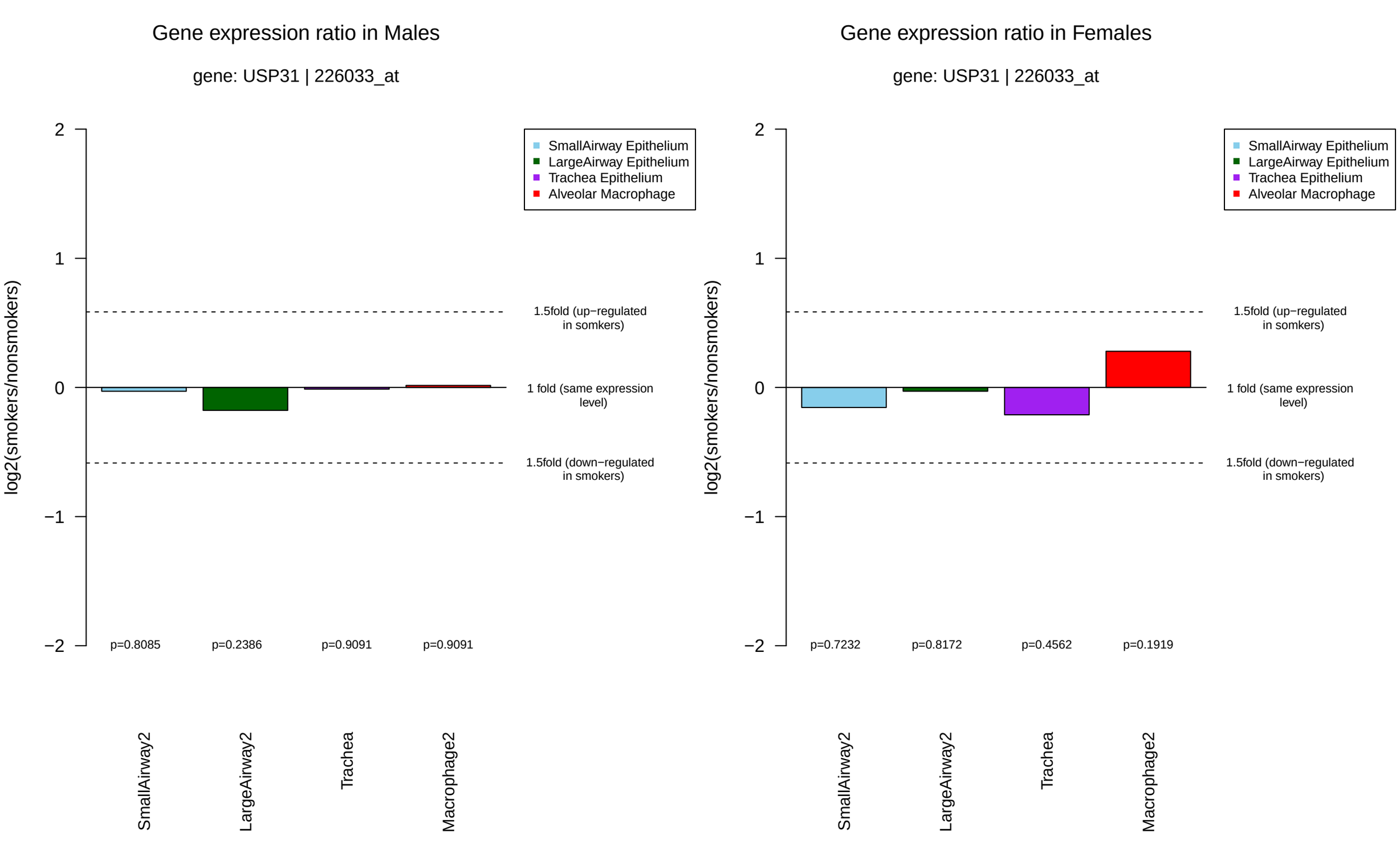
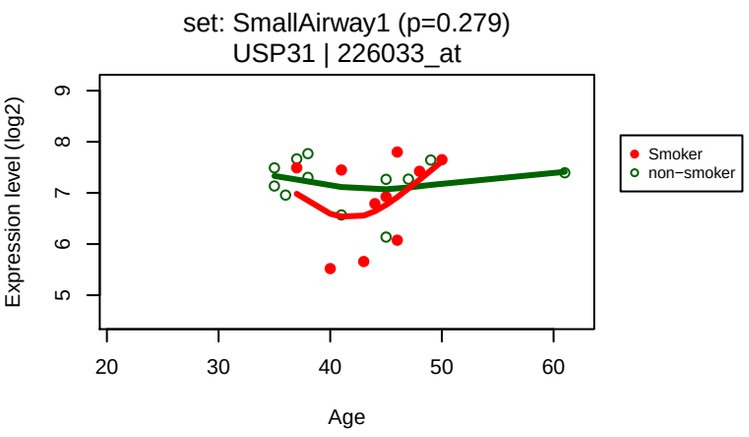
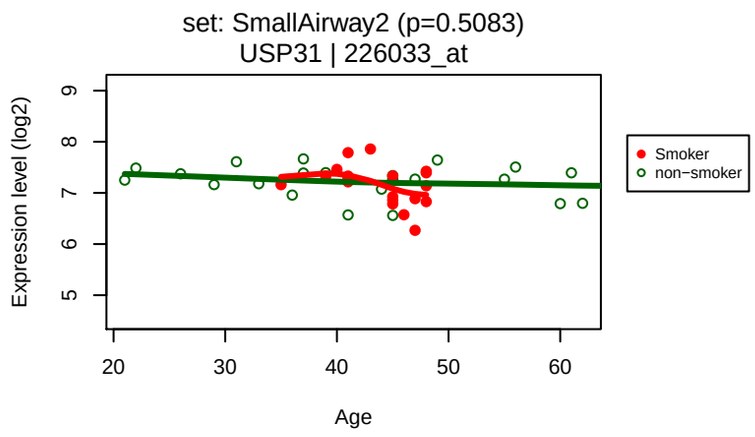
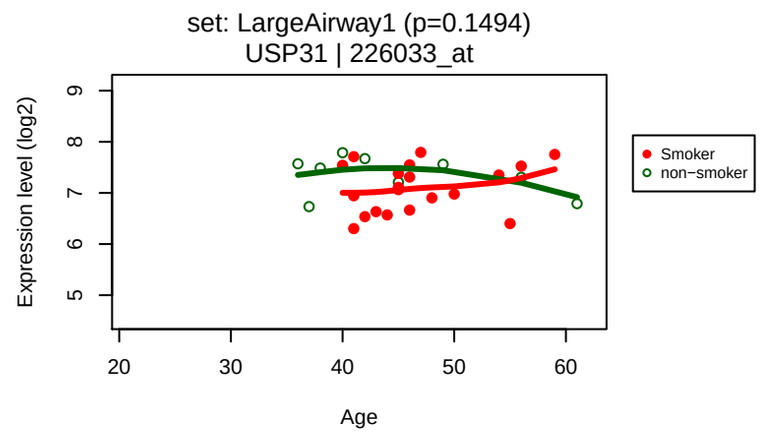

Gene expression of smokers and nonsmokers
gene: USP31 | 226033_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.279
p=0.5083
p=0.5056
p=0.1494
p=0.188
p=0.5365
p=0.1791
p=0.3977
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

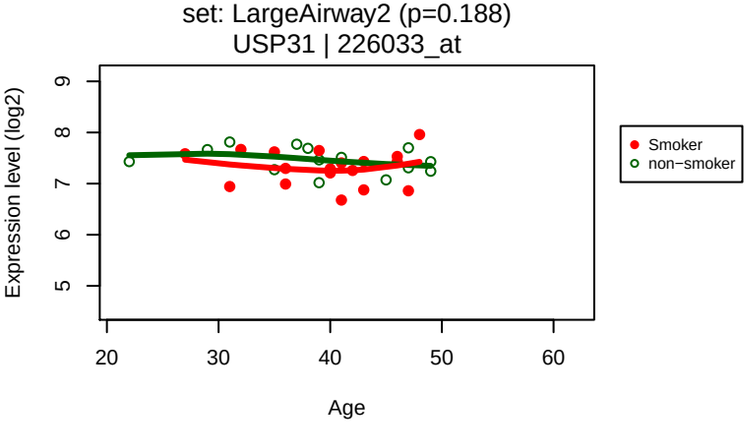

Gene expression ratio (smokers/nonsmokers)
gene: USP31 | 226033_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.279
p=0.5083
p=0.5056
p=0.1494
p=0.188
p=0.5365
p=0.1791
p=0.3977
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

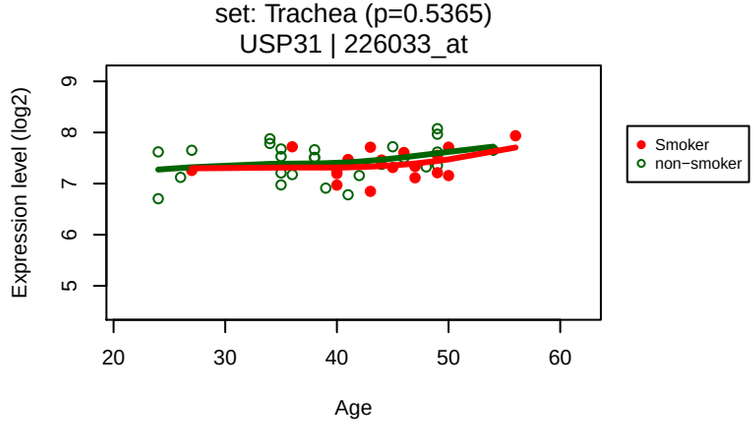

Gene expression ratio in Males
gene: USP31 | 226033_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8085
p=0.2386
p=0.9091
p=0.9091
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP31 | 226033_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7232
p=0.8172
p=0.4562
p=0.1919
SmallAirway2
LargeAirway2
Trachea
Macrophage2

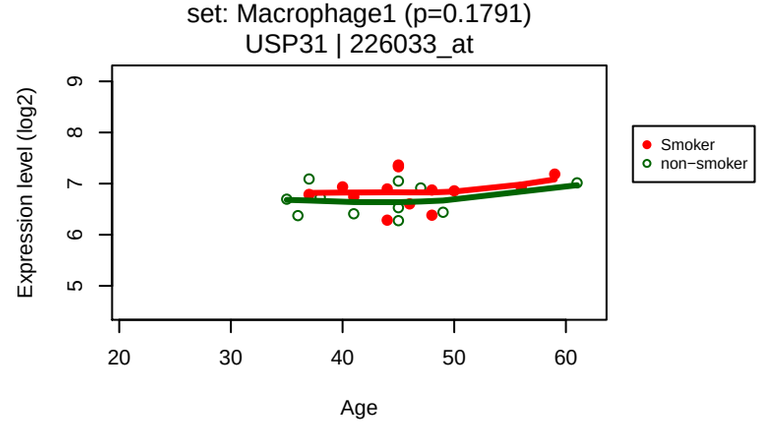

set: SmallAirway1 (p=0.279)
USP31 | 226033_at
Expression level (log2)
Age
Smoker
non−smoker

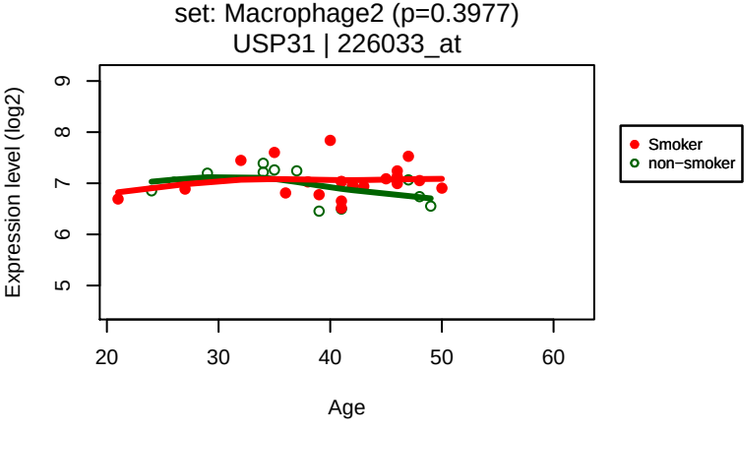

set: SmallAirway2 (p=0.5083)
USP31 | 226033_at
Expression level (log2)
Age
Smoker
non−smoker

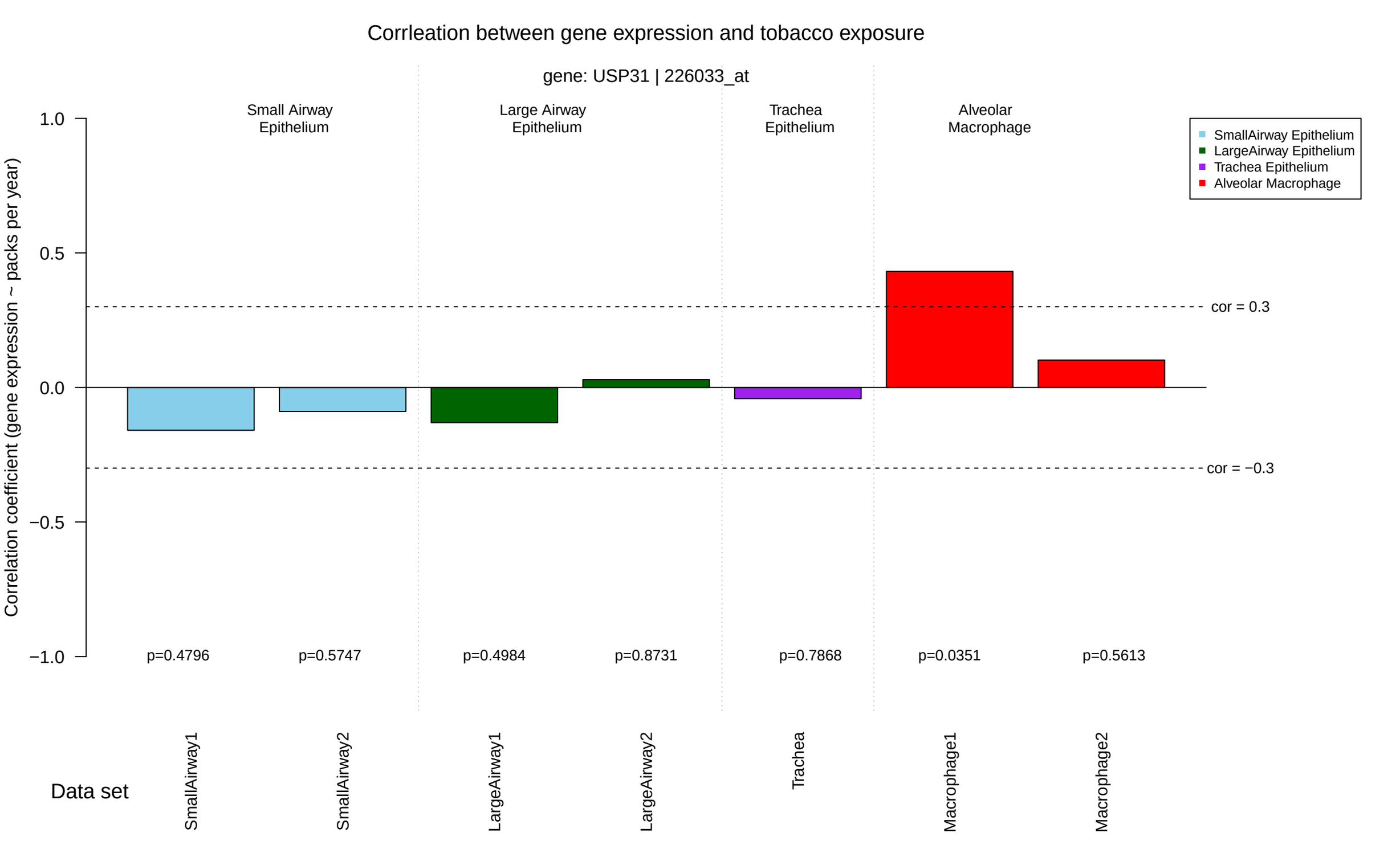

set: LargeAirway1 (p=0.1494)
USP31 | 226033_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.188)
USP31 | 226033_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5365)
USP31 | 226033_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1791)
USP31 | 226033_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3977)
USP31 | 226033_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP31 | 226033_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4796
p=0.5747
p=0.4984
p=0.8731
p=0.7868
p=0.0351
p=0.5613
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2